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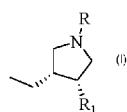
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(54) Title: PYRROLIDINE COMPOUNDS, ITS SALT AND USE IN THE PREPARATION OF UPADACITINIB THEREOF



(57) Abstract: The present invention relates to process for the preparation of a pyrrolidine compounds useful as key intermediate for the preparation of upadacitinib. More specifically the present invention relates to a process for preparing compound of Formula I, or pharmaceutically acceptable salts, polymorphs, isomers thereof. The present invention further provides a process for the preparation of upadacitinib using compounds of Formula I. Moreover, the present invention provides various crystalline solvates and crystalline hydrates of Upadacitinib or its pharmaceutically acceptable salts thereof.

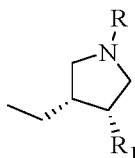
WO 2021/005484 A1

PYRROLIDINE COMPOUNDS, ITS SALT AND USE IN THE PREPARATION OF UPADACITINIB THEREOF

FIELD OF THE INVENTION

5

The present invention relates to novel compounds of Formula I or pharmaceutically acceptable salts, polymorphs, isomers thereof. Further, the present invention relates to use of compounds of Formula I for the preparation of Upadacitinib and pharmaceutically acceptable salts thereof,



Formula I

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Particularly, the present invention relates to a process for the preparation of Upadacitinib and its pharmaceutically acceptable salts by using pyrrolidine compounds of Formula I or its pharmaceutically acceptable salts.

BACKGROUND OF THE INVENTION

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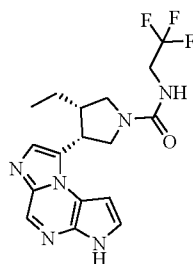
Janus kinase inhibitors, also known as JAK inhibitors or Jakinibs, are a type of medication that functions by inhibiting the activity of one or more of the Janus kinase family of enzymes (JAK1, JAK2, JAK3, TYK2), thereby interfering with the JAK-STAT signalling pathway.

20

As per references known in the art, upadacitinib is a JAK1 selective inhibitor being investigated to treat rheumatoid arthritis, Crohn's disease, ulcerative colitis, atopic dermatitis, psoriatic arthritis, axial SpA and Giant Cell Arteritis.

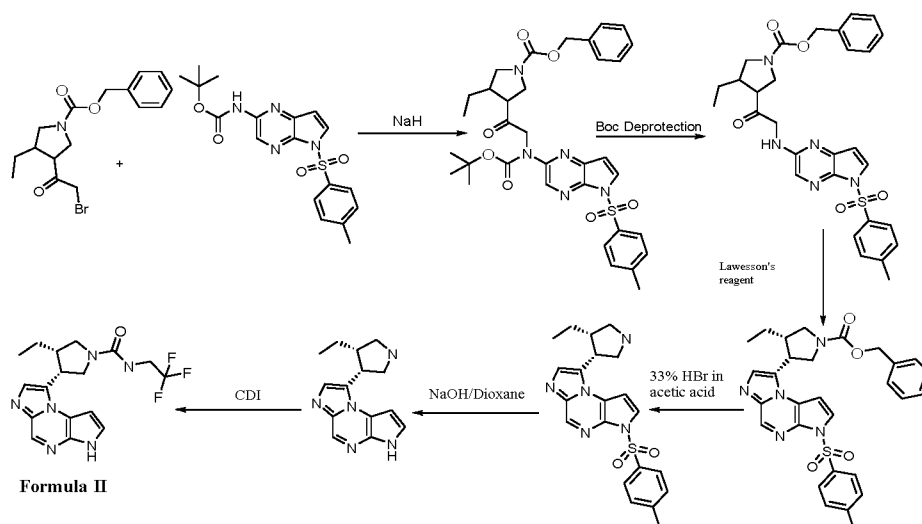
Upadacitinib chemically known as (3S,4R)-3-Ethyl-4-(3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide, is represented by structural Formula II,

25

**Formula II**

US patent number 8,426,411 B2 discloses a general process of preparing Upadacitinib. Scheme-1 mentioned below illustrates the method of the preparation of Upadacitinib as disclosed in US'411.

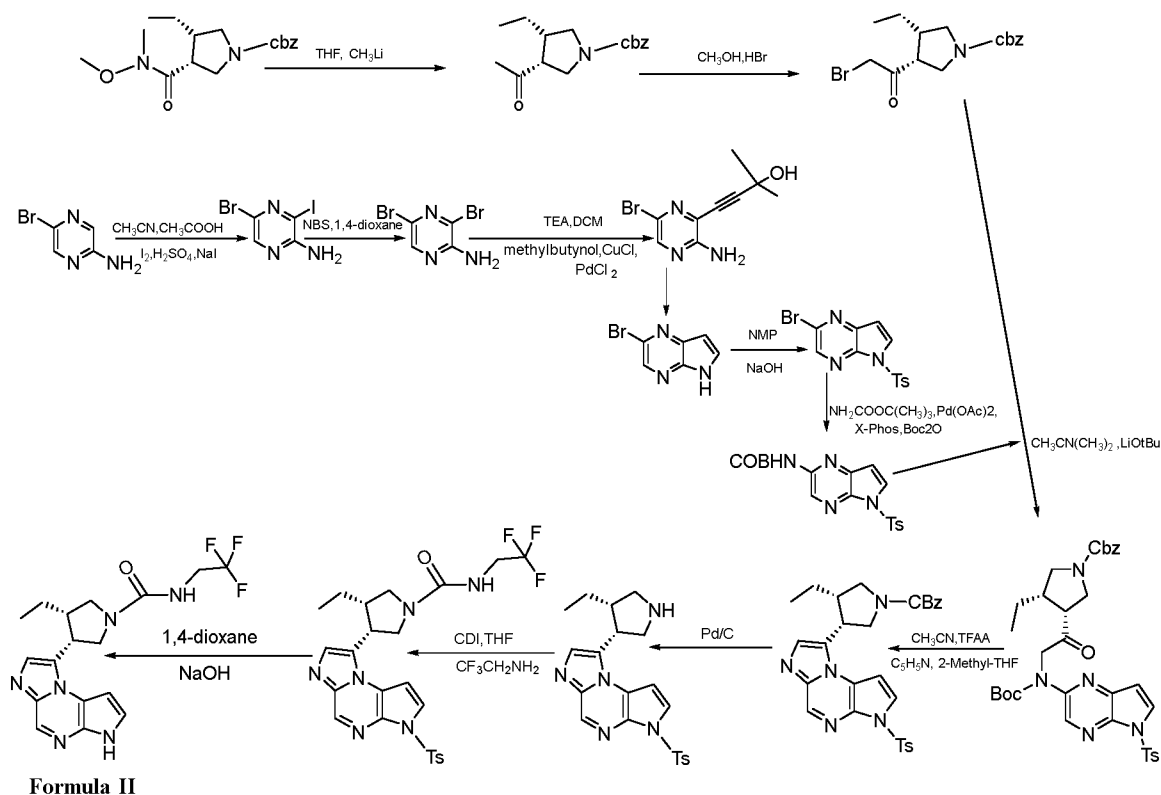
5 **Scheme-1:**

**Formula II**

PCT application number 2019/016745 A1, discloses a process of preparing (3R,4S)-1-benzyl-4-ethylpyrrolidine-3-carboxylic acid of formula II reacting pent-2-ynoic acid or its derivative thereof with an optically active sultam compound to obtain pent-2-enamide and cyclizing the pent-2-enamide compound followed by hydrolysis to obtain (3R,4S)-1-benzyl-4-ethylpyrrolidine-3-carboxylic acid.

PCT application number 2020/043033 A1, discloses process of preparing upadacitinib and its key intermediate. The process disclosed in this patent is depicted below under scheme-

2. **Scheme 2:**

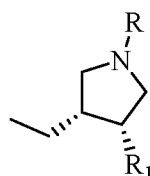


Although there are several processes known in the prior published references, however, there is remains a need for the alternate process for the preparation of novel derivatives and its salt thereof in a cost effective way.

OBJECT OF THE INVENTION

The main object of the present invention is to develop a process for the preparation of pyrrolidine compounds of Formula I or pharmaceutically acceptable salts, polymorphs, isomers thereof.

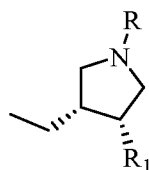
Another object of the present invention is to prepare upadacitinib of Formula II and pharmaceutically acceptable salts thereof by using pyrrolidine compounds of Formula I or pharmaceutically acceptable salts, isomers, polymorphs thereof;



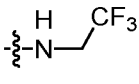
Formula I

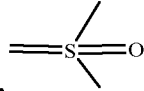
SUMMARY OF THE INVENTION

In main aspect, the present invention relates to novel compounds of Formula I, or
 5 pharmaceutically acceptable salts, polymorphs, isomers thereof;

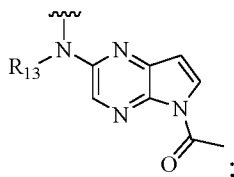
**Formula I**,

Wherein;

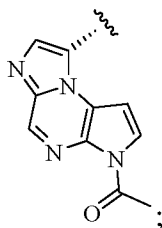
R is selected from H, -COR₂, wherein; R₂ is either  or OR₆, wherein R₆ is selected from hydrogen, (un) substituted alkyl, and (un) substituted aryl,

10 R₁ is selected from, R_b, R_c, -COR₃'', wherein; R₃'' is selected from methyl, , formyl, -NR₉OR₁₀, hydroxyl, haloalkyl, -OR₆, and -NR₄R₅, wherein; R₄ and R₅ are independently selected from hydrogen, (un) substituted alkyl, R₆ is selected from hydrogen, (un) substituted alkyl, (un) substituted aryl;

R_b is selected from -COCH₂R₁₁, and R₁₁ is represented as:



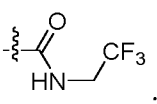
15 R_c is selected from group represented as:



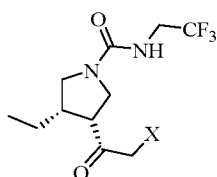
R₁₃ are independently selected from hydrogen, carbon containing moieties, silyl containing moieties and sulfur containing moieties; and

20 R₉ and R₁₀ represents alkyl group,

Provided that when R₁ is COR₃" with R₃" representing haloalkyl or hydroxyl, then R is selected

from hydrogen or .

In another aspect, the present invention provides compound of Formula I represented
5 by compound of Formula IX, pharmaceutically acceptable salts, polymorphs, isomers thereof,

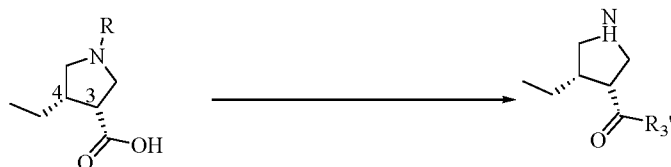


Formula IX ,

wherein, X is halogen.

In another aspect, the present invention provides a process for the preparation of (3R,
10 4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof, wherein said process comprising the steps of:

a) converting (3R,4S)-N-Protected-4-ethylpyrrolidine-3-carboxylic acid compound of Formula VI to (3R,4S)-4-ethylpyrrolidine-3-carboxylic acid of Formula VII or its salt,

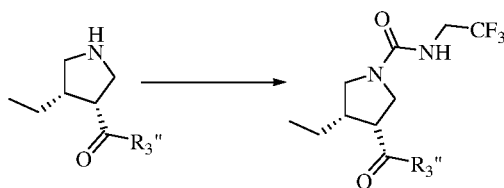


Formula VI

Formula VII ;

wherein, R and R₃" are as defined above;

b) reacting compound of Formula VII or its salt with 2,2,2-trifluoroethyl amine in presence of coupling agent to give compound of Formula VIII or its salt,

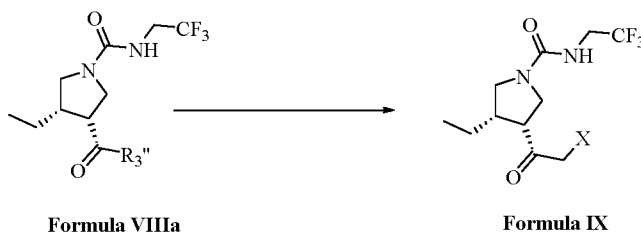


Formula VII

Formula VIII ;

20 wherein, R₃" is as defined above; and

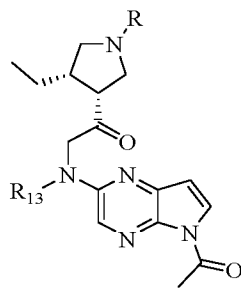
c) converting compound of Formula VIII or its salt to (3R,4S)-3-(2-haloacetyl)-4-ethyl-*N*-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof,



wherein, X is halogen; and R₃ is as defined above.

5

In one another aspect, the present invention provides compound of Formula I, represented by compound of Formula III,

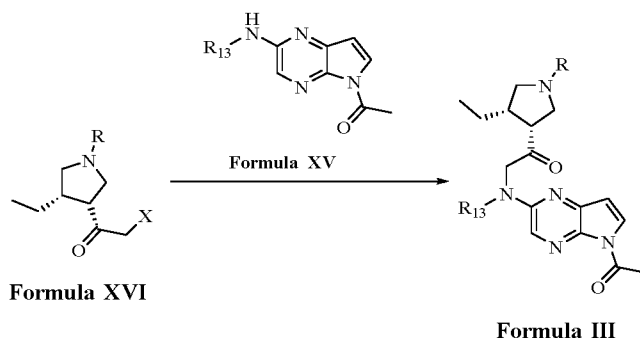


wherein,

10 R is selected from H, -COR₂, wherein; R₂ is either or OR₆, wherein R₆ is selected from hydrogen, (un) substituted alkyl, and (un) substituted aryl, R₁₃ is selected from hydrogen, carbon containing moieties, silyl containing moieties and sulfur containing moieties.

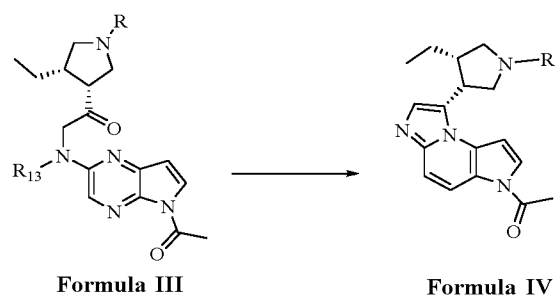
15 In another aspect, the present invention provides a process for the preparation of upadacitinib of Formula II and its pharmaceutically acceptable salts, wherein said process comprising the steps of:

a) condensing compound of Formula XVI with compound of Formula XV in presence of suitable solvent to give compound of Formula III,



wherein X, R, R₁₃ are as defined above;

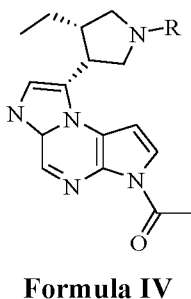
b) cyclizing compound of Formula III in presence of suitable solvent to give compound of Formula IV;



wherein R, R₁₃, are as defined above; and

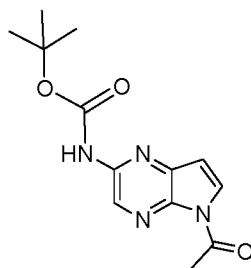
c) converting compound of Formula IV to upadacitinib of Formula II and its salt.

In another aspect, the present invention provides compounds of Formula I represented as compound of Formula IV,



wherein,

R is selected from H, -COR₂, wherein; R₂ is either or OR₆, wherein R₆ is selected from hydrogen, (un) substituted alkyl, and (un) substituted aryl.



Formula XVa

In another aspect, the present invention provides crystalline form of upadacitinib wherein said crystalline form is isolated in form of a solvate or hydrate.

5

In another aspect, the present invention provides a pharmaceutical composition comprising crystalline solvate or hydrate of upadacitinib along with atleast one pharmaceutically acceptable excipients.

10 **DETAILED DESCRIPTION**

Definitions:

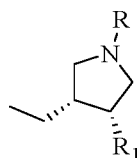
“Pharmaceutically acceptable salts” or “salts” as used in the context of the present invention refers to inorganic acids such as hydrochloric acid, hydrobromic acid, sulphuric acid, phosphoric acid salt; organic acids such as succinic acid, formic acids, acetic acid, diphenyl acetic acid, triphenylacetic acid, caprylic acid, dichloroacetic acid, trifluoro acetic acid, propionic acid, butyric acid, lactic acid, citric acid, gluconic acid, mandelic acid, tartaric acid, malic acid, adipic acid, aspartic acid, fumaric acid, glutamic acid, maleic acid, malonic acid, benzoic acid, *p*-chlorobenzoic acid, dibenzoyl tartaric acid, oxalic acid, nicotinic acid, *o*-hydroxybenzoic acid, *p*-hydroxybenzoic acid, 1-hydroxy-naphthalene-2-carboxylic acid, hydroxynaphthalene-2-carboxylic acid, ethanesulfonic acid, ethane-1,2-disulfonic acid, 2-hydroxyethane sulfonic acid, methanesulfonic acid, (+)-camphor-10-sulfonic acid, benzenesulfonic acid, naphthalene-2-sulfonic acid, *p*-toluenesulfonic acid and the like.

25 The term "excipient" or "pharmaceutically acceptable excipient" or “carrier” means a component of a pharmaceutical product that is not an active ingredient, and includes but not limited to filler, diluent, disintegrants, glidants, stabilizers, surface active agents etc. The excipients that are useful in preparing a pharmaceutical composition are generally safe, non-

toxic and neither biologically nor otherwise undesirable, and are acceptable for veterinary use as well as human pharmaceutical use. One excipient can perform more than one function.

The term “suitable solvent” as used in present invention is selected from the group comprising of alcohol such as methanol, ethanol, 2-nitroethanol, 2-fluoroethanol, 2,2,2-trifluoroethanol, hexafluoroisopropyl alcohol, ethylene glycol, 1-propanol, 2-propanol (isopropyl alcohol), 2-methoxyethanol, 1-butanol, 2-butanol, t-butyl alcohol, 2-ethoxyethanol, diethylene glycol, 1-, 2-, or 3-pentanol, neo-pentyl alcohol, t-pentyl alcohol, diethylene glycol monomethyl ether, diethylene glycol monoethyl ether, cyclohexanol, phenol, glycerol; halogenated solvent such as dichloromethane, chloroform, carbon tetrachloride, chlorobenzene; ethers such as diethyl ether, diisopropyl ether, methyl t-butyl ether, glyme, diglyme, tetrahydrofuran, 2-methyltetrahydrofuran, 1,4-dioxane, dibutyl ether, dimethylfuran, 2-methoxyethanol, 2-ethoxyethanol, anisole; ketones such as acetone, ethyl methyl ketone, diethyl ketone, methyl isobutyl ketone; esters solvents such as ethyl acetate, n-propyl acetate, n-butyl acetate, iso propyl acetate, isobutyl acetate, t-butyl acetate, ethyl formate, methyl acetate, methyl propanoate, ethyl propanoate, methyl butanoate, ethyl butanoate; hydrocarbon such as toluene, xylene, hexane, n-heptane, n-pentane, anisole, ethyl benzene and the like; nitriles such as acetonitrile, propionitrile, butanenitrile; formamides such as dimethyl formamide; acetamides such as dimethyl acetamide; pyrrolidines such as N-methyl pyrrolidine; morpholine; pyridine; sulfoxides such as dimethyl sulfoxide; carbonates; water; and mixtures thereof.

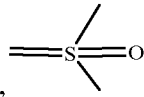
In one embodiment, the present invention provides novel compounds of Formula I, or pharmaceutically acceptable salts, polymorphs, isomers thereof,



Formula I,

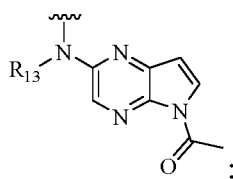
Wherein;

R is selected from H, -COR₂, wherein; R₂ is either $\frac{1}{2}\text{-N}(\text{H})(\text{CF}_3)$ or OR₆, wherein R₆ is selected from hydrogen, (un) substituted alkyl, and (un) substituted aryl,

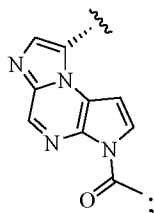
R_1 is selected from, Rb, Rc, $-\text{COR}_3''$, wherein; R_3'' is selected from methyl, , formyl, $-\text{NR}_9\text{OR}_{10}$, hydroxyl, haloalkyl, $-\text{OR}_6$, and $-\text{NR}_4\text{R}_5$,

wherein; R_4 and R_5 are independently selected from hydrogen, (un) substituted alkyl, R_6 is selected from hydrogen, (un) substituted alkyl, (un) substituted aryl;

- 5 Rb is selected from $-\text{COCH}_2\text{R}_{11}$, and R_{11} is represented as:



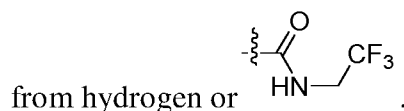
Rc is selected from group represented as:



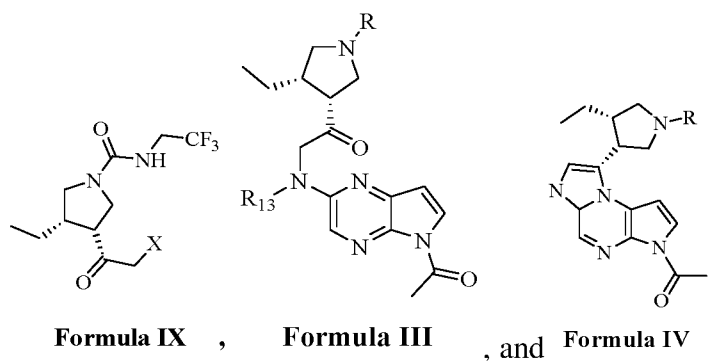
- 10 R_{13} are independently selected from hydrogen, carbon containing moieties, silyl containing moieties and sulfur containing moieties; and

R_9 and R_{10} represents alkyl group,

Provided that when R_1 is COR_3'' with R_3'' representing haloalkyl or hydroxyl, then R is selected

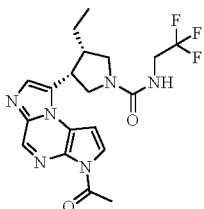
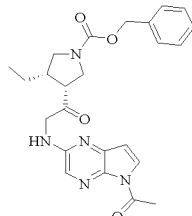
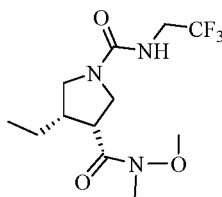
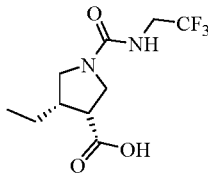
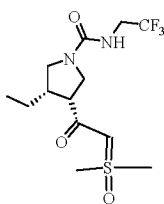
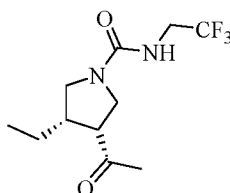
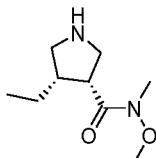


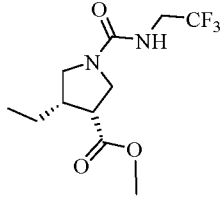
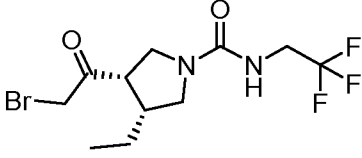
- 15 In another embodiment, the present invention provides compound of Formula I represented by compounds of Formulae IX, III and IV,



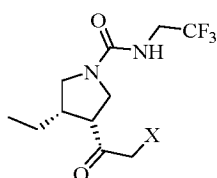
wherein, X, R and R_{13} are as defined above.

In another embodiment, the present invention provides compounds of Formula I selected from the compounds represented as follows:

Sr. No	Chemical Name	Sr. No	Chemical Structure
1.	(3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-c]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide	2.	benzyl (3R,4S)-3-((5-acetyl-5H-pyrrolo [2,3-b]pyrazin-2-yl)glycyl)-4-ethylpyrrolidine-1-carboxylate
			
3.	(3R,4S)-4-ethyl-N3-methoxy-N3-methyl-N1-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide	4.	(3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid
			
5.	(3R,4S)-3-(2-(dimethyl(oxo)-16-sulfanylidene)acetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide	6.	(3R,4S)-3-acetyl-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide
			
7.	(3R,4S)-4-ethyl-N-methoxy-N-methylpyrrolidine-3-carboxamide	8.	methyl (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylate
			

			
9.	(3R, 4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide		
			

In another embodiment, the present invention provides compound of Formula I represented by compound of Formula IX, pharmaceutically acceptable salts, polymorphs, isomers thereof,



Formula IX ,

wherein, X is halogen.

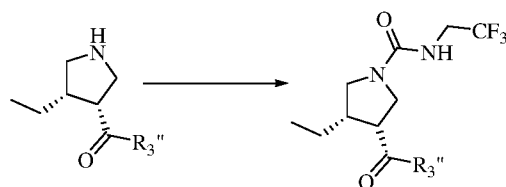
In another embodiment, the present invention provides a process for the preparation of (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof, wherein said process comprising the steps of:

a) converting (3R,4S)-N-Protected-4-ethylpyrrolidine-3-carboxylic acid compound of Formula VI to compound of Formula VII,



wherein, R and R₃" are as defined above;

b) reacting compound of Formula VII with 2,2,2-trifluoroethyl amine in presence of coupling agent to give compound of Formula VIIIa or its salt,



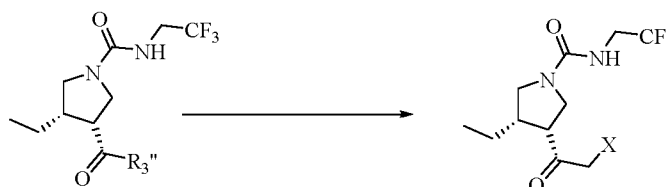
Formula VII

Formula VIIIa

;

wherein, R_3'' is as defined above; and

c) converting compound of Formula VIIIa or its salt to (3R,4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof,



Formula VIIIa

Formula IX

,

wherein, X is halogen; and R_3'' is as defined above.

In another embodiment, the present invention provides a process for the preparation of (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide compound of Formula IX or its salts, wherein said process further comprising the steps of:

a) converting compound of Formula VIIIa to (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid of Formula X in a suitable solvent,



Formula VIIIa

Formula X

,

wherein, R_6 is as defined above;

b) amidating (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid compound of Formula X to give (3R,4S)-4-ethyl-N³-methoxy-N³-methyl-N1-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide of Formula XI;

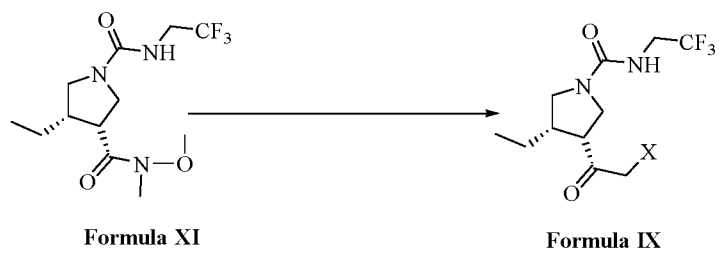


Formula X

Formula XI

; and

c) converting compound of Formula XI to (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide compound of Formula IX or its salts,

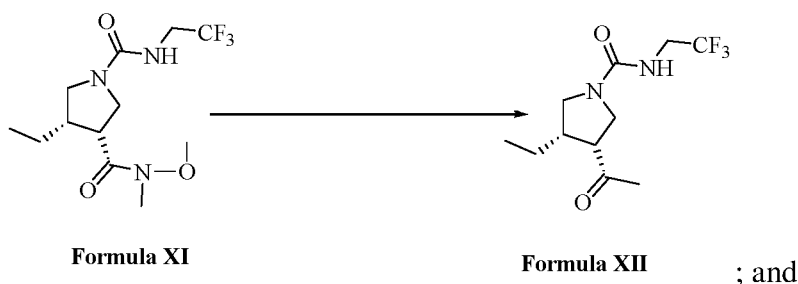


wherein, X is halogen.

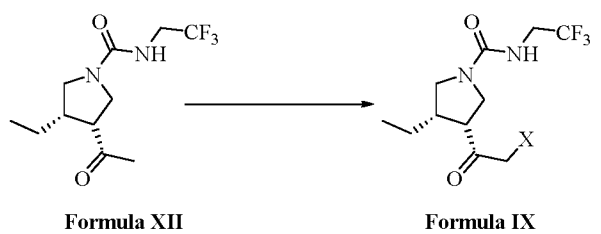
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In another embodiment, the present invention provides a process for the preparation of (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide compound of Formula IX or its pharmaceutically acceptable salts, wherein said process comprising the steps of:

10 a) converting compound of Formula XI to compound of Formula XII,



b) halogenating compound of Formula XII to give compound of Formula IX or its salts,



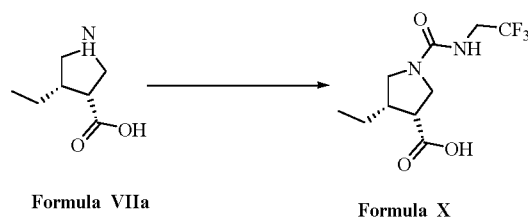
15 wherein, X is halogen.

In another embodiment, the present invention provides process for the preparation of upadacitinib of Formula II and its salts by using compound of Formula IX or salts, isomers, polymorphs thereof.

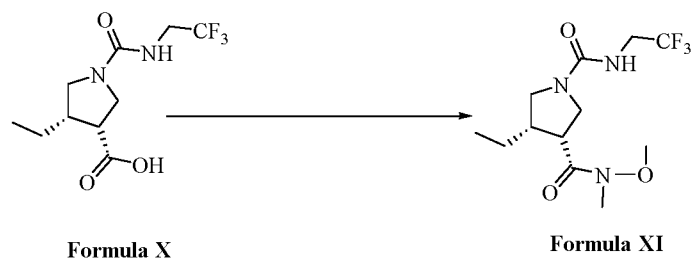
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In another embodiment, the present invention provides a process for the preparation of upadacitinib of Formula II and its salts by using (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide compound of Formula IX or its salts, wherein said process comprising the steps of:

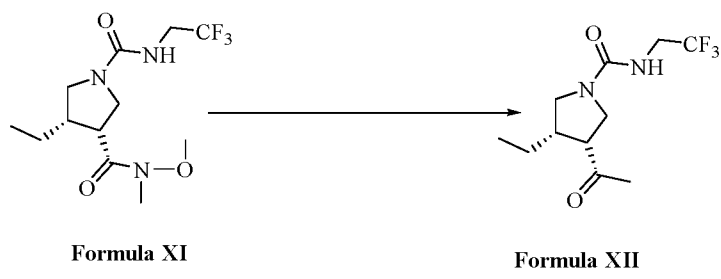
- 5 a) converting compound of Formula VIIa to compound of Formula X in a suitable solvent and coupling agent,



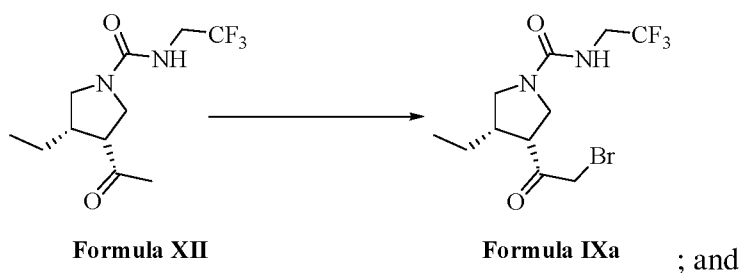
- b) amidating (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid compound of Formula X to give (3R,4S)-4-ethyl-N³-methoxy-N³-methyl-N1-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide of Formula XI;
- 10



- c) converting compound of Formula XI to (3R,4S)-3-acetyl-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula XII,



- 15 d) brominating compound of Formula XII to get compound of Formula IXa,



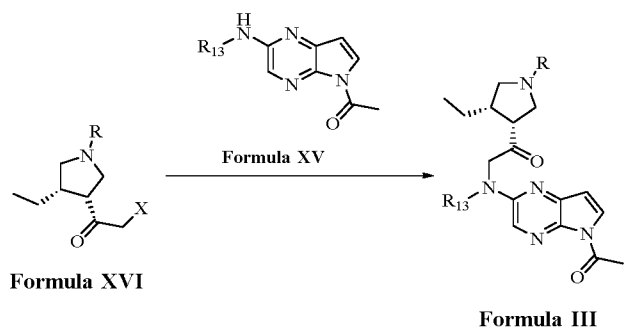
e) converting compound of Formula IXa to upadacitinib of Formula II and its pharmaceutically acceptable salt.

In an embodiment, suitable coupling agent used in present invention is selected from the group consisting of carbonyldiimidazole (CDI), carbonyl-di(1,2,4-triazole), 1-ethyl-3-(3-dimethylamino propyl)carbodiimide (EDC), dicyclohexylcarbodiimide (DCC), 1,3-diisopropyl carbodiimide (DIC), 1-ethyl-3-(3-dimethyl aminopropyl)carbodiimide hydrochloride (WSC.HCl), and Hydroxybenzotriazole (HOBT).

In another embodiment, the bromination of ((3R,4S)-3-acetyl-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide is performed in presence of a bromine source and an alcoholic solvent, wherein said brominating agent is selected from N-bromosuccinimide or elemental bromine

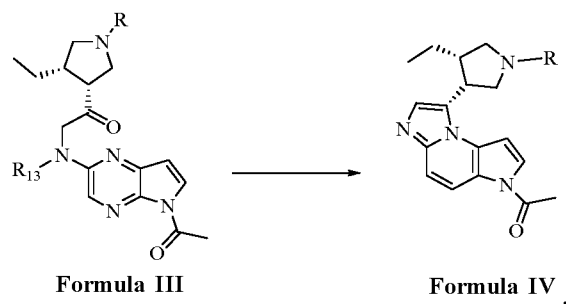
In another embodiment, the present invention provides process for the preparation of upadacitinib of Formula II and its pharmaceutically acceptable salts, wherein said process comprising the steps of:

a) condensing compound of Formula XVI with compound of Formula XV in presence of suitable solvent to give compound of Formula III,



wherein X, R, R₁₃, are as defined above;

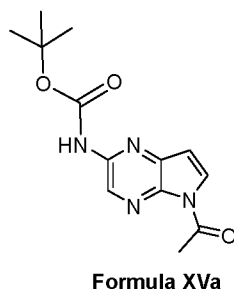
b) cyclizing compound of Formula III in presence of suitable solvent to give compound of Formula IV;



wherein R, R₁₃, are as defined above; and

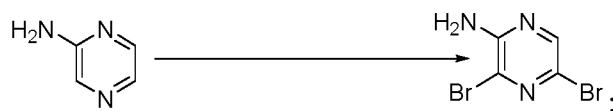
c) converting compound of Formula IV to upadacitinib of Formula II and its salt.

- 5 In another embodiment, the present invention provides process for the preparation of compound of Formula XV, wherein R₁₃ represent Boc group and the compound is represented as compound of Formula XVa,

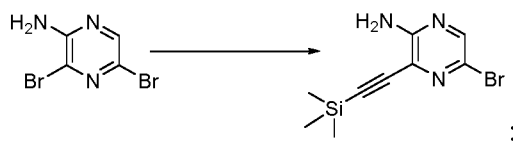


wherein said process comprising the steps of:

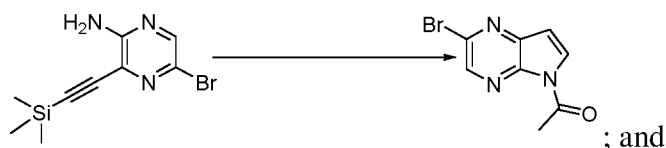
- 10 a) brominating pyrazine-2-amine in presence of a brominating agent and solvent,



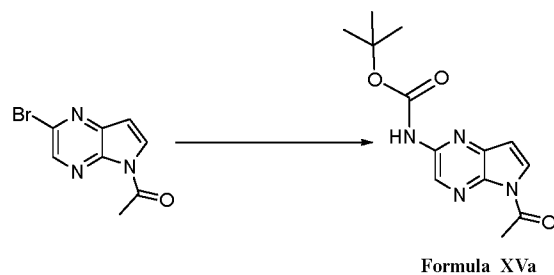
b) reacting 3,5-dibromopyrazine-2-amine obtained in previous step with trimethylsilylacetylene in presence of palladium catalyst to give 5-bromo-3-((trimethylsilyl)ethynyl)pyrazin-2-amine,



- 15 c) cyclizing and acetylating 5-bromo-3-((trimethylsilyl)ethynyl)pyrazin-2-amine in presence of base to give 1-(2-bromo-5H-pyrrolo[2,3-b]pyrazin-5-yl)ethanone,

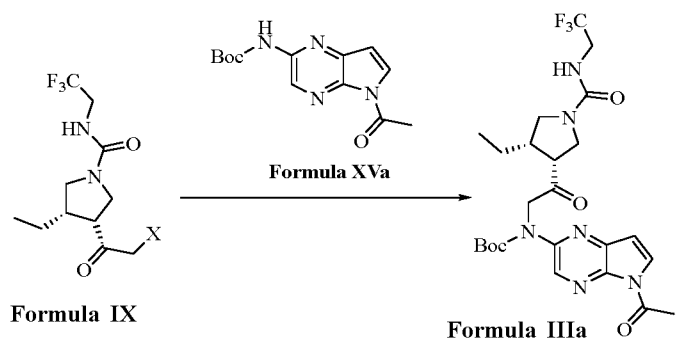


d) reacting 1-(2-bromo-5H-pyrrolo[2,3-b]pyrazin-5-yl)ethanone with tert-butyl carbamate to give tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate or its salt.

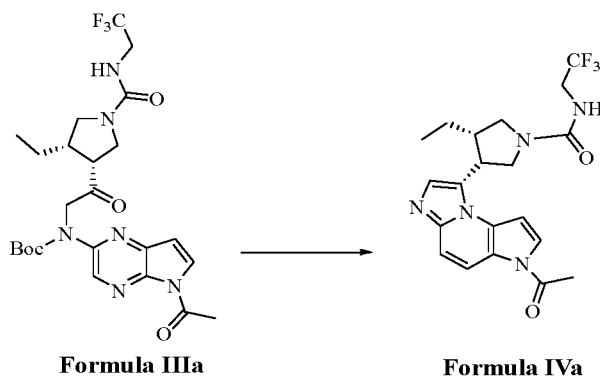


5 In a specific embodiment, the present invention provides a process of preparing upadacitinib or pharmaceutically acceptable salt thereof, wherein said process comprising the steps of:

a) reacting (3R,4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof, with tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate or its salt of Formula XVa, to give compound of Formula IIIa,



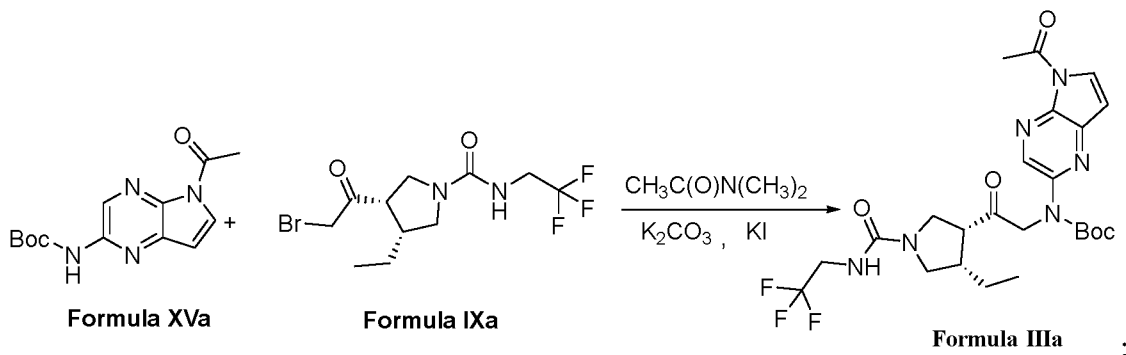
b) cyclizing compound of Formula IIIa in presence of suitable solvent to give (3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IVa;



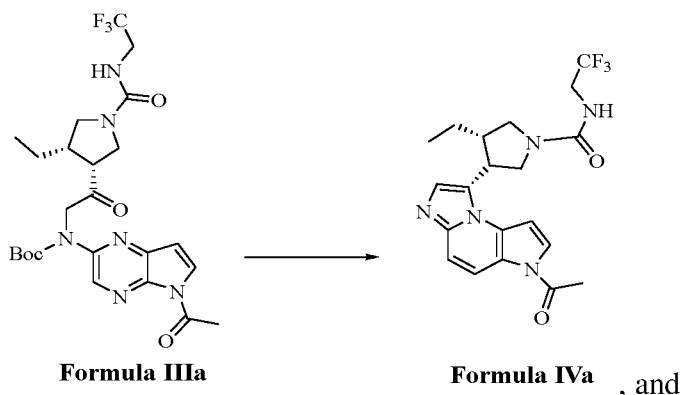
c) converting to upadacitinib or its pharmaceutically acceptable salt.

In another specific embodiment, the upadacitinib of Formula II or its salt is prepared in following steps:

a) reacting tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate of Formula XVa with (3R,4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)-pyrrolidine-1-carboxamide of Formula IXa in presence of inorganic base to give tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)(2-((3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidin-3-yl)-2-oxoethyl)carbamate of Formula IIIa,



b) cyclizing tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)(2-((3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidin-3-yl)-2-oxoethyl)carbamate in presence of Lawesson's reagent to give (3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IVa;



c) converting to upadacitinib or its pharmaceutically acceptable salt.

In an embodiment, reaction between compound of Formula IXa and Formula XVa is carried out in presence, suitable base selected from the group comprising of metal carbonate such as lithium carbonate, sodium carbonate, potassium carbonate, barium carbonate, calcium carbonate and magnesium carbonate; metal bicarbonate such as sodium bicarbonate, potassium bicarbonate, barium bicarbonate, calcium bicarbonate and magnesium bicarbonate and metal

hydroxide such as sodium hydroxide, potassium hydroxide, barium hydroxide, calcium hydroxide magnesium hydroxide and lithium tert-butoxide.

In another embodiment, the present invention provides process for the purification of upadacitinib or its pharmaceutically acceptable salt, comprising the steps of:

a) adding upadacitinib free base or pharmaceutically acceptable salt thereof in a suitable solvent to get a reaction mass;

b) optionally heating the reaction mass; and

c) removing the solvent to get pure upadacitinib free base or pharmaceutically acceptable salt thereof.

In another embodiment, the present invention provides process for the purification of upadacitinib or its salt, comprising the steps of:

a) dissolving upadacitinib free base in a suitable solvent and treating with suitable acid or base to form upadacitinib salt;

b) neutralizing or desalting the upadacitinib salt to give upadacitinib free base;

c) optionally converting the upadacitinib free base to its pharmaceutically acceptable salt;

d) providing a solution of upadacitinib free base of step b) or its salt of step c) in a suitable solvent (s); and

e) removing the solvent from the solution obtained in step e) to get pure upadacitinib or its salt.

In another embodiment, the present invention provides process of preparing crystalline solvate of upadacitinib, wherein said process comprising the steps of:

a) dissolving upadacitinib of Formula II in a solvent to get a solution;

b) optionally, cooling to a suitable temperature to get crystals; and

c) isolating the crystals from step a) or step b) to get crystalline solvate of upadacitinib of Formula II.

In another embodiment, the present invention further provides crystalline form of upadacitinib hydrate selected from monohydrate, dihydrate, trihydrate, hemihydrate and sesquihydrate of upadacitinib.

In another preferred embodiment, upadacitinib solvates prepared according to present invention may include, but not limited to, methanol solvate, ethanol solvate, 1-Propanol solvate, 2-Propanol solvate, butyl acetate solvate, isoamyl acetate solvate, n-propyl acetate solvate, dimethyl sulfoxide solvate, ethyl acetate solvate, methyl acetate solvate, acetone solvate, methyl ethyl ketone solvate, methyl isobutyl ketone solvate, acetonitrile solvate and the like.

In another embodiment, the present invention provides crystalline form of upadacitinib wherein said crystalline form is optionally isolated as anhydrous in nature.

In a preferred embodiment, the present invention provides Upadacitinib pharmaceutically acceptable salts wherein said salts are selected from, but not limited to, fumarate salt, oxalate salt, tosylate salt, mesylate salt, citrate salt, mandelate salt, succinate acetate salt, diphenyl acetate salt, triphenylacetic acid salt, caprylic acid salt, dichloroacetic acid salt, trifluoro acetic acid salt, propionate salt, butyrate salt, lactate, gluconic acid salt, tartarate salt, maleate salt, adipate salt, aspartate salt, glutamate salt, malonic acid salt, benzoate salt, *p*-chlorobenzoate salt, dibenzoyl tartarate salt, nicotinic acid salt, *o*-hydroxybenzoic acid salt, *p*-hydroxybenzoic acid salt, 1-hydroxy-naphthalene-2-carboxylic acid salt, hydroxynaphthalene-2-carboxylic acid salt, ethanesulfonic acid salt, ethane-1,2-disulfonic acid salt, 2-hydroxyethane sulfonic acid salt, methanesulfonic acid salt, (+)-camphor-10-sulfonic acid salt, benzenesulfonic acid salt, naphthalene-2-sulfonic acid salt, *p*-toluenesulfonic acid salt, hydrobromide salt, and hydrochloride salt

In further embodiment, the present invention provides upadacitinib or its pharmaceutically acceptable salt characterized by particle size distribution wherein, d_{90} is between 0.1 μ m to 200 μ m, specifically d_{90} is between 2.0 μ m to 150 μ m.

In one another embodiment, the upadacitinib and its pharmaceutically acceptable salt prepared as per the process of the present invention is characterized with purity above 99%, preferably above 99.5%, and more preferably above 99.9%.

In one another embodiment, the present invention further provides a composition comprising upadacitinib or its pharmaceutically acceptable salt and one or more

pharmaceutical acceptable excipients, wherein said upadacitinib or its pharmaceutically acceptable salt is prepared as per the process of the present invention.

EXAMPLES

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Example 1: Synthesis of (3R, 4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide

Step-1: Synthesis of (3R, 4S)-4-ethyl-1-((2, 2, 2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid

- 10 Charged (3R, 4S)-4-ethylpyrrolidine-3-carboxylic acid (10 g) in tetrahydrofuran (50 ml) in a 250 ml four neck round bottom flask. A separate solution of 1,1-carbonyldiimidazole (17 gm), tetrahydrofuran (50 ml) and 2,2,2-trifluoroethyl amine 10.39 g) was prepared in 100 ml round bottom flask at 20-25°C. Charged the later to the former solution and stirred for 4-5 hours and after completion of reaction, quenched the reaction mass with 20% citric acid solution (50ml).
- 15 Reaction mass obtained after quenching was stirred for 1 hour and extracted the compound in 50 ml of ethyl acetate at 20-25°C. Separated the organic layer and washed with brine. Distilled the organic layer under vacuum at 50-55°C to get 18g of the title product.

Step-2: Synthesis of (3R, 4S)-4-ethyl-N³-methoxy-N³-methyl-N¹-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide

- 20 A mixture of (3R, 4S)-4-ethyl-1-((2, 2, 2-trifluoroethyl)carbamoyl)pyrrolidine-3- carboxylic acid (18g), EDC. HCl (16.1g), dimethylolpropionic acid DMPA (24.88g) and N,O-dimethylhydroxylamine hydrochloride in dichloromethane (270 ml) was prepared in a round bottom flask at 20-25°C and stirred the reaction mass for 2-3 hours. Then the reaction mixture
- 25 was quenched with the water at 20-25°C and stirred for 15 min to separate the organic layer. Then the organic layer was distilled out to get 20.0g of the title product.

Step-3: Synthesis of (3R, 4S)-3-acetyl-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide

- 30 (3R,4S)-4-ethyl-N³-methoxy-N³-methyl-N¹-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide (20 g) was charged in tetrahydrofuran (200ml) in a round bottom flask at 20-25°C under nitrogen atmosphere and mixture was cooled to 0-5°C and was slowly added methyl magnesium bromide in ether (3M) (63.5ml) at 0-5°C. Stirred the reaction mass for 1 hr at 20-25°C. Saturated ammonium chloride solution (200 ml) and ethyl acetate (100ml) was added at

20-25°C and the reaction mass was stirred for 10-15 minute at 20-25°C. Separated the organic layer and concentrated completely to get 17g of the title product.

Step-4: Synthesis of (3R, 4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl) pyrrolidine-1-carboxamide

(3R, 4S)-3-acetyl-4-ethyl-N-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide (17g) and bromine (15.3g) was charged in methanol (170ml) at 0-25°C and stirred it for 4-5 hours at 20-25°C. Cooled the reaction mass to 0-5°C and added sodium bisulphite (90ml). Ethyl acetate (95ml) was charged at 20-25°C and stirred to separate the organic layer. Organic layer was distilled out completely at 40-45°C to get the title compound.

Example 2: Synthesis of tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate

Step 1: Synthesis of 3,5-dibromopyrazin-2-amine

Charged 10.0g of 2-aminopiprazine in 100ml of dichloromethane, cooled to 0-5 °C and added 39.3g of NBS lot wise and stirred for 2-3 hrs at 0-5°C. After completion of reaction, quenched the reaction mass with sodium bi-sulfite solution. Separated the organic layer and degassed at 40-45°C to get the title compound.

Step 2: Synthesis of 5-Bromo-3-((trimethylsilyl)acetylene pyrazine-2-amine

15.0 g of 3,5-dibromopyrazin-2-amine was charged in THF (150.0ml) followed by addition of CuI (1.13g) and TEA(8.0ml). Added 3.25g of Pd(PPh₃)₄ and cooled to 0-5°C under continuous stirring. Added 6.4g pf Trimethylsilylacetylene and stirred for 2-3 hrs at 0-5°C. Extracted the compound with ethyl acetate and separated the organic layer. Distilled the organic layer at 40-45°C and dried the solid so obtained to get the title compound.

Step 3: 1-(2- Bromo 5-H-Pyrrolo[2,3-b] pyrazine-5-yl) ethanone

15.0 g of 5-Bromo-3-((trimethylsilyl)acetylene pyrazine-2-amine was dissolved in THF (150.0 ml) and sodium hydride(1.46g), at 0-5° and stirred for 0.5 hrs at 0-5°C. Added 5.66g of acetyl chloride at 0-5°C and stirred overnight. After completion of reaction, quenched the reaction mixture with chilled water and extracted the product with ethyl acetate. Separated the organic layer and concentrated under reduced pressure to get 12.0 g of the title compound.

Step 4: Synthesis of tert-Butyl 5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-ylcarbamate

Charged 10.0g of 1-(2-Bromo-5-H-Pyrrolo[2,3-b] pyrazine-5-yl) ethanone in 200 ml of THF and to it was added tert-butyl carbamate (4.8 g), potassium carbonate (17.2 g), palladiumacetate (0.93g) and 9,9-dimethyl-4,5-bis(diphenylphosphino)-xanthene (Xantphos) (0.48g) in a three-neck flask and refluxed overnight. Cooled the reaction mixture at RT and extracted the organic compound with ethyl acetate. Distilled the organic layer under reduced pressure to get the title compound.

Example 3: Synthesis of Upadacitinib

Step 1: Preparation of tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)(2-((3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidin-3-yl)-2-oxoethyl)carbamate

Charged tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate (5.6 g), dimethyl acetamide (50.0 ml), potassium carbonate (4.2 g), and potassium iodide (0.7 g) at 20-25°C. Stirred for 15-20 minute at 20-25°C. Charged solution of (3R,4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide (7.0 g) in dimethyl acetamide (25.0 ml) at 20-25°C. Stirred for 5-6hours at 20-25°C. Charged DM water (150.0 ml) and ethyl acetate (70.0 ml) at 20-25°C. Stirred it for 20-30 minute at 20-25°C. Separated the layers and distilled the organic layer and degassed to get 10.1 g of desired compound.

Step 2: Preparation of (3R,4S)-3-((5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)glycyl)-4-ethyl-N-(2,2,2 trifluoroethyl)pyrrolidine-1-carboxamide

Charged tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)(2-((3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidin-3-yl)-2-oxoethyl)carbamate to dichloromethane (100.0 ml) at 20-25°C. Charged trifluoroacetic acid (17.0 g) at 0-5°C. Allowed to increase the temperature to 20-25°C. Stirred for 15-20 hours at 20-25°C. Charged saturated sodium bicarbonate solution (200.0 ml) at 20-25°C. Stirred for 25-30 minute at 20-25°C. Separated the layers and distilled the organic layer to get 6.1 of desired compound.

Step 3: Preparation of (3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide

(3R,4S)-3-((5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)glycyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide, 1,4-dioxan (30.0 ml), and toluene (120.0 ml) in a 500 ml four neck round bottom flask at 20-25°C. Purged argon gas through reaction mass for 25-30 minute at 20-25°C. Charged Lawesson's reagent (4.13g) at 20-25°C. Heated the reaction mass to 80°C. Stirred the reaction mass for 4-5 hour at 80.0°C under argon atmosphere. After completion of

reaction, charged sat. sodium bicarbonate solution (100.0ml) and ethyl acetate (60.0ml) at 20-25°C. Stirred and separated the layer at 20-25°C. Distilled out the organic layer under vacuum and degassed to get 4.3 g of desired compound.

5 **Step 4: Synthesis of Upadacitinib**

Charged (3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide (4.0 g), to solution containing DM water (4.0ml) and sodium hydroxide (2.6 g) to 1,4-dioxan (40.0ml) at 20-25°C. Heated to 50-60°C and stirred for 3-4 hours. Cooled the reaction mass to 20-25°C. Extracted the compound with 40.0 ml of ethyl acetate at 20-25°C. Separated the organic layer at 20-25°C. Washed the organic layer with 20.0 ml brine solution at 20-25°C. Separated the organic layer at 20-25°C. Distilled out the organic layer under vacuums at 50-55°C and crystallized with methanol and water to get 3.1g of desired compound with HPLC purity of 99.8%

OR

Charged (3R,4S)-3-((5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)glycyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide to 1,4-dioxan (30.0 ml), and toluene (120.0 ml) in a 500 ml four neck round bottom flask at 20-25°C. Purged argon gas through reaction mass for 25-30 minute at 20-25°C. Charged Lawesson's reagent (4.13g) at 20-25°C. Heated the reaction mass to 80°C. Stirred the reaction mass for 4-5 hour at 80.0°C under argon atmosphere. Cooled the reaction mass to 20-25°C. After 4-5 hours, added 1,4-dioxan (40.0ml), DM water (4.0ml) and potassium hydroxide (2.6 g) at 20-25°C. Heated it to 50-60°C. Stirred it for 3-4 hours. Cooled it to 20-25°C. Extracted with 40.0 ml ethyl acetate at 20-25°C. Separated the organic layer at 20-25°C. Washed the organic layer with 20.0 ml of brine solution at 20-25°C. Separated the organic layer at 20-25°C. Distilled out the organic layer under vacuums at 50-55°C and crystallized with methanol and water to get 3.1g of desired compound with HPLC purity of 99.9%

Example 4: Preparation of Upadacitinib n-butyl acetate solvate

Charged 0.2 mL of DM water to solution of Upadacitinib free base (500 mg) dissolved in n-butyl acetate (2.5 mL) at 20-25 °C, and stirred for 10-15 minute at 20-25°C. Cooled the final solution to -16 °C to -20 °C for 48 hour. The mass obtained was filtered under dried vacuum at -16 °C to -20 °C. Air dried the material to get 425.0 mg of desired compound.

Example 5: Preparation of Upadacitinib n-propyl acetate solvate

Took 0.2 mL DM water to a 10 mL vial. Upadacitinib free base (500 mg) was dissolved in n-propyl acetate (2.5 mL) at 20-25°C and stirred for 10-15 minute at 20-25 °C. Upadacitinib solution was added slowly to vial containing water with stirring. Cooled the final solution to -16 °C to -20 °C for 48 hour. The mass obtained was filtered under vacuum at -16 °C to -20 °C.

5 Air dried the material to get 470.0 mg of desired compound.

Example 6: Preparation of Upadacitinib isoamyl acetate solvate

Took 0.2 mL DM water in a 10 mL vial. Upadacitinib free base (500 mg) was dissolved in isoamyl acetate (2.5 mL) at 20-25 °C and stirred for 10-15 minute at 20-25 °C. Upadacitinib solution was added slowly to vial containing water with stirring. Cooled the final solution to -16 °C to -20 °C for 48 hour. The mass obtained was filtered under vacuum at -16 °C to -20 °C. Air dried the material to get 430.0 mg of desired compound.

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Example 7: Preparation of Upadacitinib 1-Propanol solvate

Charged 0.2 mL DM water in a 10 mL vial. Upadacitinib free base (500 mg) was dissolved in 1-Propanol (2.5 mL) at 20-25 °C and stirred for 10-15 minute at 20-25°C. Upadacitinib solution was added slowly to vial containing water with stirring. Cooled the final solution to -16 °C to -20 °C for 48 hour. The mass obtained was filtered under vacuum at -16 °C to -20 °C. Air dried the material to get 400.0 mg of desired compound.

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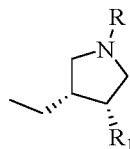
Example 8: Preparation of Upadacitinib ethyl acetate solvate

Upadacitinib free base (500 mg) was dissolved in ethyl acetate (2.5 mL) at 20-25 °C and stirred for 10-15 minute at 20-25°C. Upadacitinib solution was added slowly to vial containing 0.2 ml of water with stirring. Cooled the final solution to -16 °C to -20 °C for 48 hour. The mass obtained was filtered under vacuum at -16 °C to -20 °C. Air dried the material to get 490.0 mg of desired compound.

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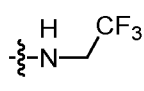
CLAIMS:

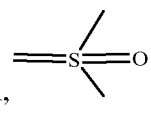
1. A compound of Formula I or pharmaceutically acceptable salts, polymorphs, isomers thereof,



Formula I,

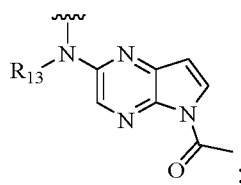
wherein;

R is selected from H, -COR₂, wherein; R₂ is either  or OR₆, wherein R₆ is selected from hydrogen, (un) substituted alkyl, and (un) substituted aryl,

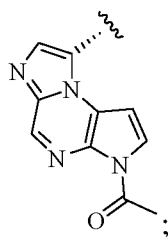
R₁ is selected from, R_b, R_c, -COR₃'', wherein; R₃'' is selected from methyl, , formyl, -NR₉OR₁₀, hydroxyl, haloalkyl, -OR₆, and -NR₄R₅,

wherein; R₄ and R₅ are independently selected from hydrogen, (un) substituted alkyl, R₆ is selected from hydrogen, (un) substituted alkyl, (un) substituted aryl;

R_b is selected from -COCH₂R₁₁, and R₁₁ is represented as:



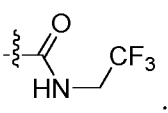
R_c is selected from group represented as:



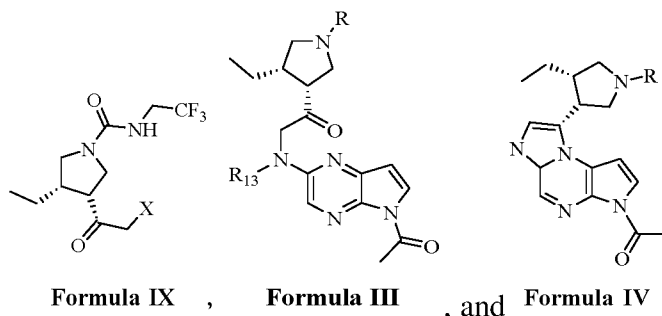
R₁₃ are independently selected from hydrogen, carbon containing moieties, silyl containing moieties and sulfur containing moieties; and

R₉ and R₁₀ represents alkyl group,

Provided that when R₁ is COR₃'' with R₃'' representing haloalkyl or hydroxyl, then R is selected

from hydrogen or .

2. The compounds as claimed in claim 1, wherein said compounds are selected from the following:



wherein, X, R and R₁₃ are as defined above.

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3. The compounds as claimed in claim 2, wherein said compounds are selected from the following:

(3R,4S)-3-(3-acetyl-3H-imidazo[1,2-a]pyrrolo[2,3-e]pyrazin-8-yl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide;

10 benzyl (3R,4S)-3-((5-acetyl-5H-pyrrolo [2,3-b]pyrazin-2-yl)glycyl)-4-ethylpyrrolidine-1-carboxylate;

(3R,4S)-4-ethyl-N3-methoxy-N3-methyl-N1-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide;

(3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid;

15 (3R,4S)-3-(2-(dimethyl(oxo)-16-sulfanylidene)acetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide;

(3R,4S)-3-acetyl-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide;

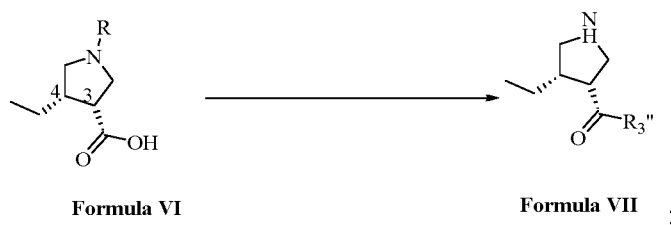
(3R,4S)-4-ethyl-N-methoxy-N-methylpyrrolidine-3-carboxamide;

methyl (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylate; and

20 (3R, 4S)-3-(2-bromoacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide

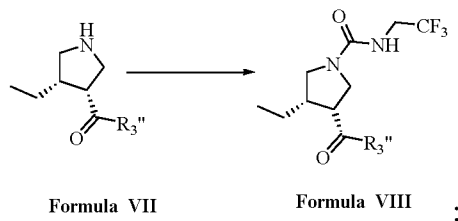
4. A process for the preparation of (3R, 4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof, wherein said process comprising the steps of:

25 a) converting (3R,4S)-N-Protected-4-ethylpyrrolidine-3-carboxylic acid compound of Formula VI to (3R,4S)-4-ethylpyrrolidine-3-carboxylic acid of Formula VII or its salt,



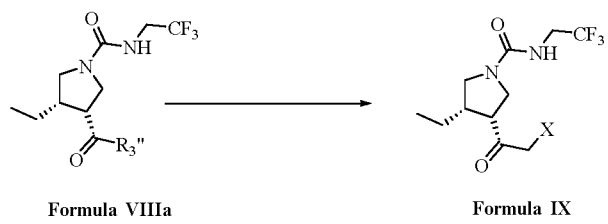
wherein, R and R₃' are as defined above;

b) reacting compound of Formula VII or its salt with 2,2,2-trifluoroethyl amine in presence of coupling agent to give compound of Formula VIII or its salt,



wherein, R₃' is as defined above; and

c) converting compound of Formula VIII or its salt to (3R,4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof,

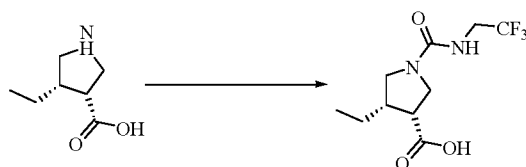


wherein, X is halogen; and R₃' is as defined above.

5. The process as claimed in claim 4, wherein said coupling agent is selected from the group consisting of carbonyldiimidazole (CDI), carbonyl-di(1,2,4-triazole), 1-ethyl-3-(3-dimethylamino propyl)carbodiimide (EDC), dicyclohexylcarbodiimide (DCC), 1,3-diisopropyl carbodiimide (DIC), 1-ethyl-3-(3-dimethyl aminopropyl)carbodiimide hydrochloride (WSC.HCl), and Hydroxybenzotriazole (HOBT).

6. The process as claimed in claim 4, wherein said process of preparing compound of Formula IX comprising the steps of:

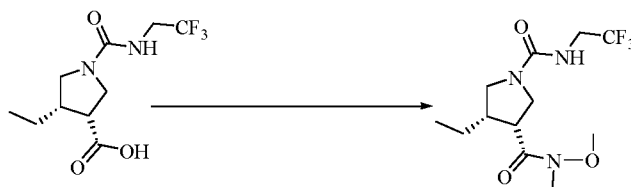
a) converting compound of Formula VIIa to compound of Formula X in a suitable solvent and coupling agent,



Formula VIIa

Formula X

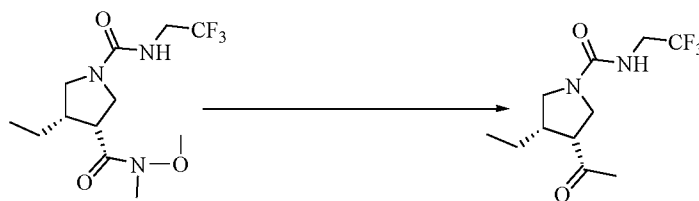
b) amidating (3R,4S)-4-ethyl-1-((2,2,2-trifluoroethyl)carbamoyl)pyrrolidine-3-carboxylic acid compound of Formula X to give (3R,4S)-4-ethyl-*N*³-methoxy-*N*³-methyl-*N*1-(2,2,2-trifluoroethyl)pyrrolidine-1,3-dicarboxamide of Formula XI;



Formula X

Formula XI

c) converting compound of Formula XI to (3R,4S)-3-acetyl-4-ethyl-*N*-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula XII,

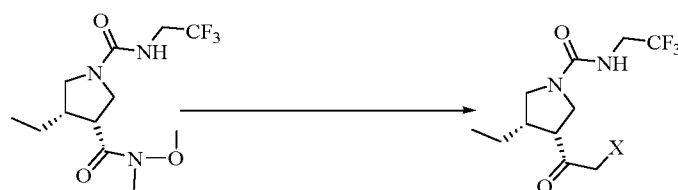


Formula XI

Formula XII

; and

d) converting compound of Formula XI to (3R, 4S)-3-(2-haloacetyl)-4-ethyl-*N*-(2, 2, 2-trifluoroethyl) pyrrolidine-1-carboxamide compound of Formula IX or its salts,



Formula XI

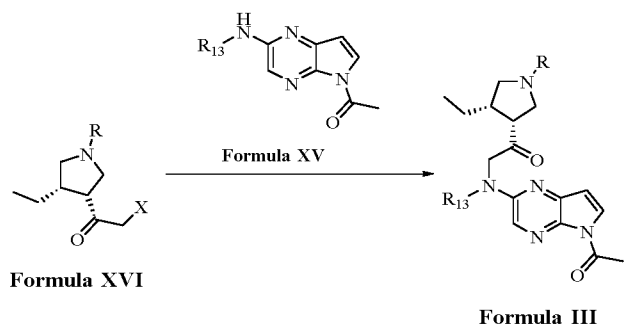
Formula IX

;

wherein, X is halogen.

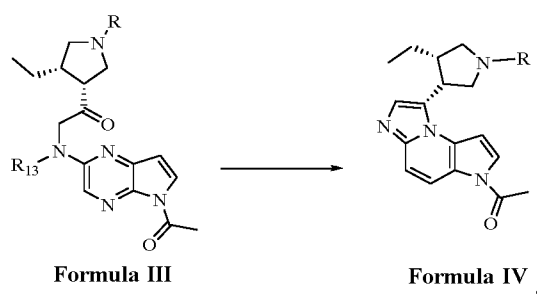
7. A process for the preparation of upadacitinib of Formula II and its pharmaceutically acceptable salts by using compounds of Formula III and IV as claimed in claim 2, wherein said process comprising the steps of:

a) condensing compound of Formula XVI with compound of Formula XV in presence of suitable solvent to give compound of Formula III,



wherein X, R, R₁₃, are as defined above;

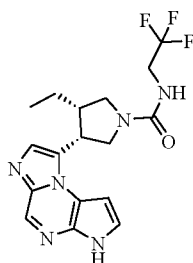
b) cyclizing compound of Formula III in presence of suitable solvent to give compound of Formula IV;



wherein R, R₁₃, are as defined above; and

c) converting compound of Formula IV to upadacitinib of Formula II and its salt.

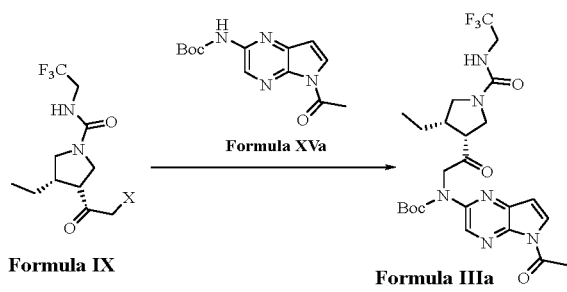
8. A process for the preparation of upadacitinib of Formula II and its pharmaceutically acceptable salts by using compounds of Formula IX as claimed in claim 2,



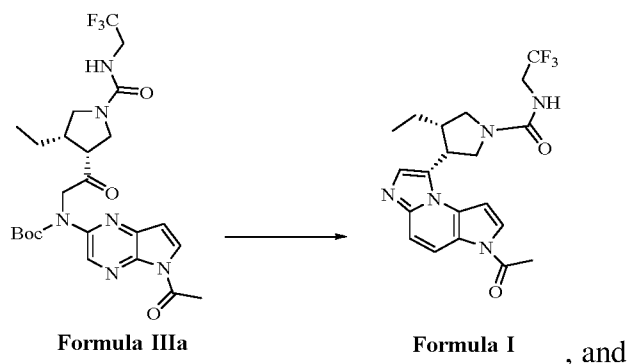
Formula II

wherein said process comprising the steps of:

a) reacting (3R,4S)-3-(2-haloacetyl)-4-ethyl-N-(2,2,2-trifluoroethyl)pyrrolidine-1-carboxamide of Formula IX or salts thereof, with tert-butyl (5-acetyl-5H-pyrrolo[2,3-b]pyrazin-2-yl)carbamate or its salt of Formula XVa, to give compound of Formula IIIa,



b) cyclizing compound of Formula IIIa in presence of suitable solvent to give upadacitinib of Formula I;



5 c) optionally purifying and converting to upadacitinib pharmaceutically acceptable salt.

9. A process for the purification of upadacitinib or its pharmaceutically acceptable salt, comprising the steps of:

a) adding upadacitinib free base or pharmaceutically acceptable salt thereof in a suitable solvent to get a reaction mass;

b) optionally heating the reaction mass; and

c) removing the solvent to get pure upadacitinib free base or pharmaceutically acceptable salt thereof.

10. A process of preparing crystalline solvate of upadacitinib, wherein said process comprising the steps of:

a) dissolving upadacitinib of Formula II in a solvent to get a solution;

b) optionally, cooling to a suitable temperature to get crystals; and

c) isolating the crystals from step a) or step b) to get crystalline solvate of upadacitinib of Formula II.

11. The process as claimed in claim 10, wherein said solvate is selected from methanol solvate, ethanol solvate, 1-Propanol solvate, 2-Propanol solvate, butyl acetate solvate, isoamyl acetate

solvate, n-propyl acetate solvate, dimethyl sulfoxide solvate, ethyl acetate solvate, methyl acetate solvate, acetone solvate, methyl ethyl ketone solvate, methyl isobutyl ketone solvate, acetonitrile solvate.

- 5 12. Crystalline Upadacitinib solvate selected from methanol solvate, ethanol solvate, 1-Propanol solvate, 2-Propanol solvate, butyl acetate solvate, isoamyl acetate solvate, n-propyl acetate solvate, dimethyl sulfoxide solvate, ethyl acetate solvate, methyl acetate solvate, acetone solvate, methyl ethyl ketone solvate, methyl isobutyl ketone solvate, acetonitrile solvate.

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13. Upadacitinib or its pharmaceutically acceptable salt as prepared by the process as claimed in any of the preceding claim, wherein said upadacitinib or its pharmaceutically acceptable salt is characterized by particle size distribution wherein, d_{90} is between $0.1\mu\text{m}$ to $200\mu\text{m}$, specifically d_{90} is between $2.0\mu\text{m}$ to $150\mu\text{m}$.

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INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2020/056321

A. CLASSIFICATION OF SUBJECT MATTER
C07D401/12 Version=2020.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

C07D

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

STN data base, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO2017066775A1 (ABBVIE INC., 20 APRIL, 2017) the whole document, particularly example 3, scheme XV, XVI	1-13
Y	US20130072470A1 (ABBVIE INC., 21 MARCH, 2013) the whole document, particularly scheme III, example 16	1-13
Y	CN109369659A (ZHEJIANG NORMAL UNIVERSITY, 22 FEBRAURY, 2019) the whole document, particularly para 009-011, 035	1-13
A	Simon boothroyd et. al. "Why Do Some Molecules Form Hydrates or Solvates?", Cryst. Growth Des. 2018, 18, 3, 1903-1908, https://doi.org/10.1021/acs.cgd.8b00160 . the whole document	10-13

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:

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"D" document cited by the applicant in the international application

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"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"I" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

15-10-2020

Date of mailing of the international search report

15-10-2020

Name and mailing address of the ISA/

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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IB2020/056321

Citation	Pub.Date	Family	Pub.Date
WO 2017066775 A1	20-04-2017	EP 3362455 A1	22-08-2018
		CA 3002220 A1	20-04-2017
		JP 2019501865 A	24-01-2019
		US 9879018 B2	30-01-2018
		CN 108368121 A	03-08-2018
		US 9951080 B2	24-04-2018
US 20130072470 A1	21-03-2013	WO 2013043826 A1	28-03-2013